LAB 3: MONITORING, LOGGING & DEVSECOPS

Task 2 — Provision Scalable Infrastructure Using Terraform

Practical Report with Execution Screenshots

Marks: 5

Objective

To provision infrastructure using Terraform (Infrastructure as Code) by creating configuration files, initializing Terraform, reviewing the execution plan, and applying the configuration to create Docker-based Nginx containers.

Tools Used

Windows PowerShell, Terraform v1.16.6, Docker Desktop, the Docker provider for Terraform, and Nginx (nginx:alpine).

Configuration Summary

The Terraform configuration defines one Docker image resource (nginx:alpine) and two Nginx containers. Container ports are mapped to host ports 8081 and 8082 so both instances can be accessed separately.

Procedure and Results

1. Install and verify Terraform

Terraform v1.16.6 was verified in Windows PowerShell.

2. Initialize and validate

The Docker provider was installed, Terraform initialization succeeded, and validation reported that the configuration is valid.

3. Plan infrastructure

Terraform proposed creating two containers (nginx-1 and nginx-2) and one image resource (nginx:alpine). Plan summary: 3 to add, 0 to change, 0 to destroy.

4. Apply configuration

After approving the plan by entering “yes”, Terraform created the image and both containers successfully.

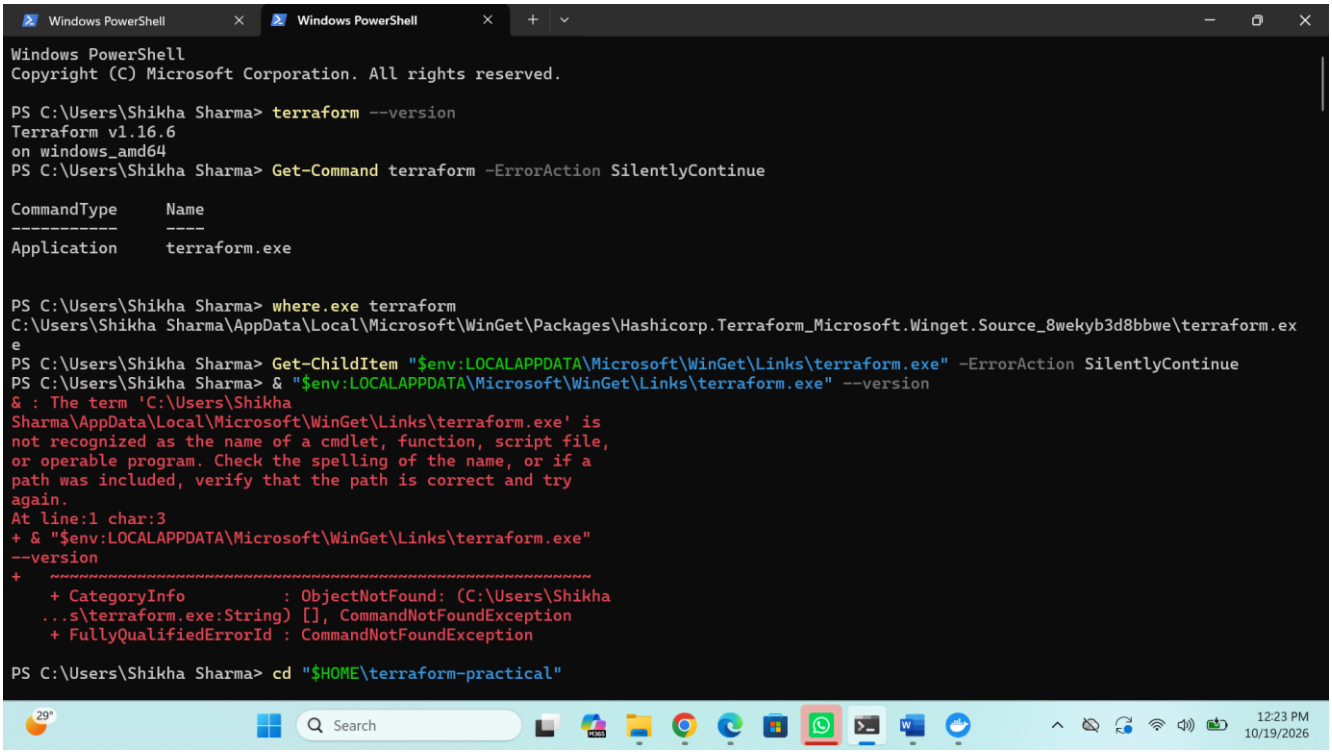
5. Verify deployment

The docker ps output shows both containers running. Host port 8081 maps to container port 80 for nginx-1, and host port 8082 maps to container port 80 for nginx-2.

Conclusion

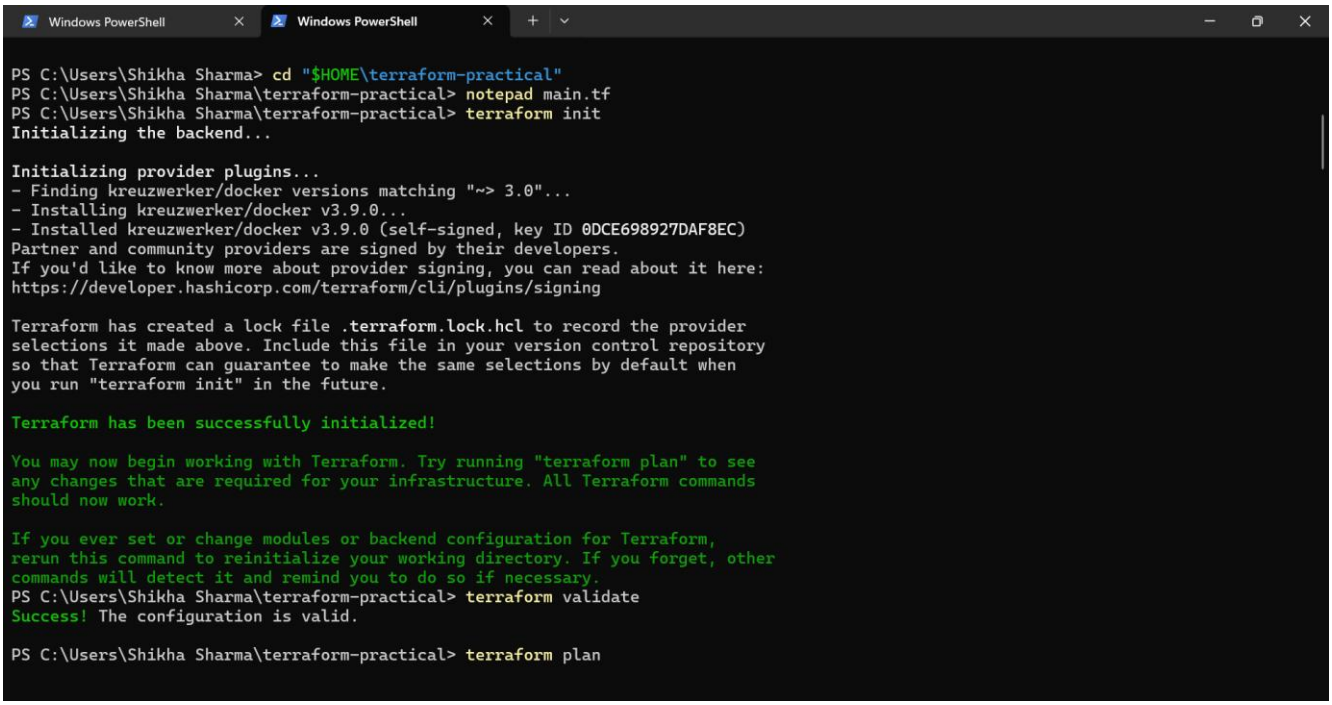
Infrastructure was provisioned successfully using Terraform as Infrastructure as Code. The final apply result was 3 resources added, 0 changed, and 0 destroyed, with both Nginx containers running.

Execution Screenshot 1



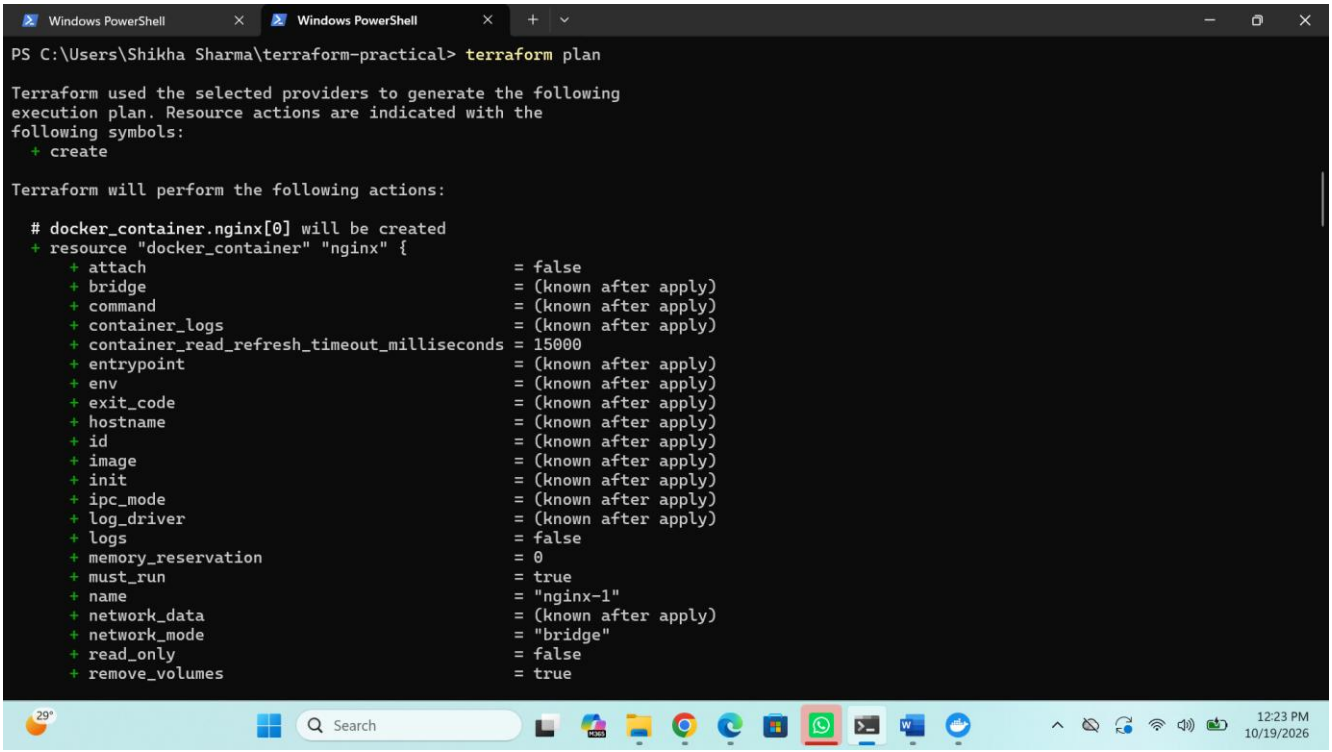
Terraform version and command-path verification.

Execution Screenshot 2



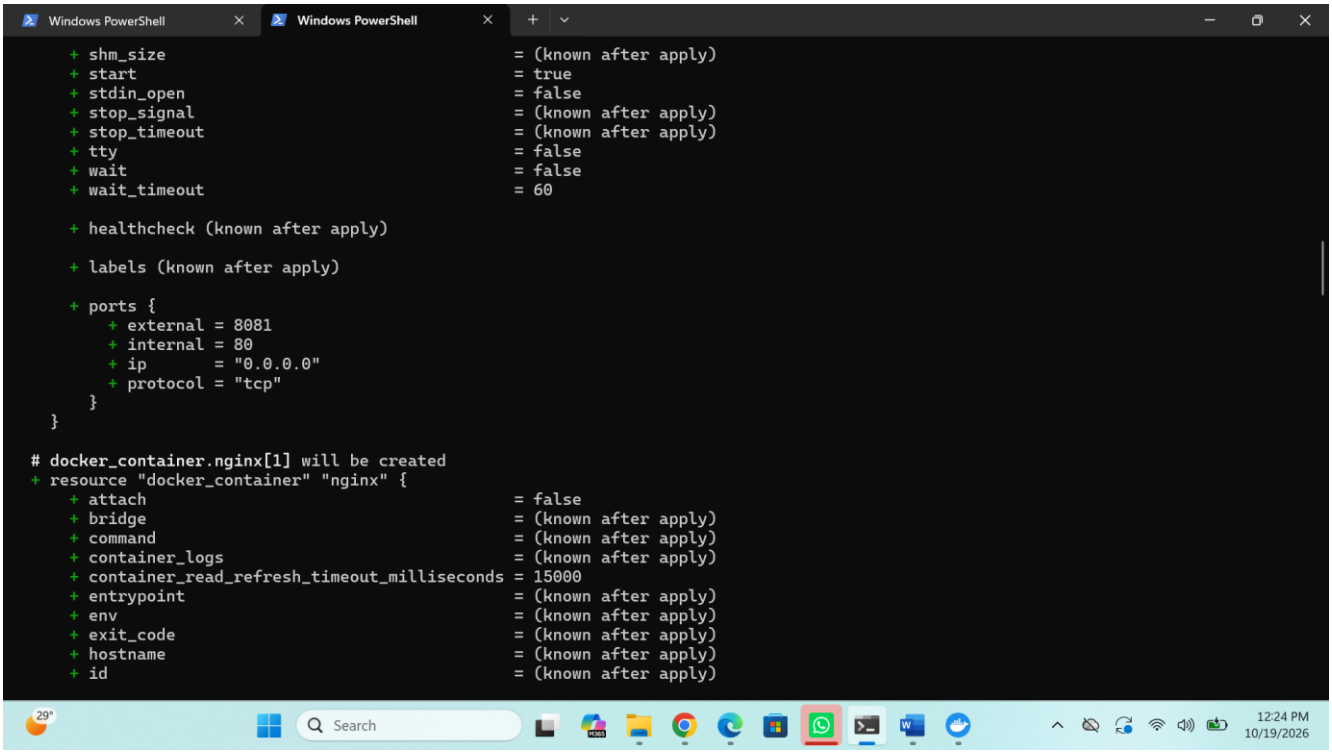
Terraform initialization and configuration validation.

Execution Screenshot 3



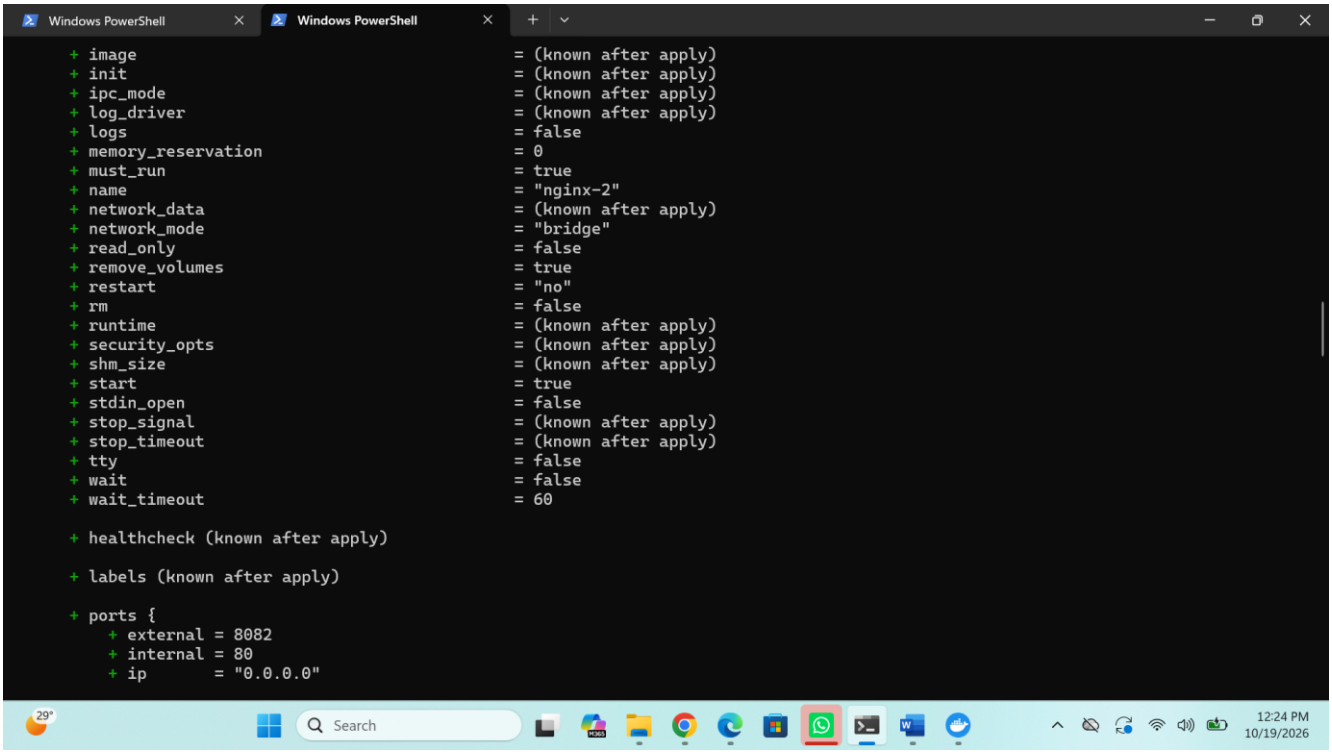
Terraform plan showing the first Nginx container resource.

Execution Screenshot 4



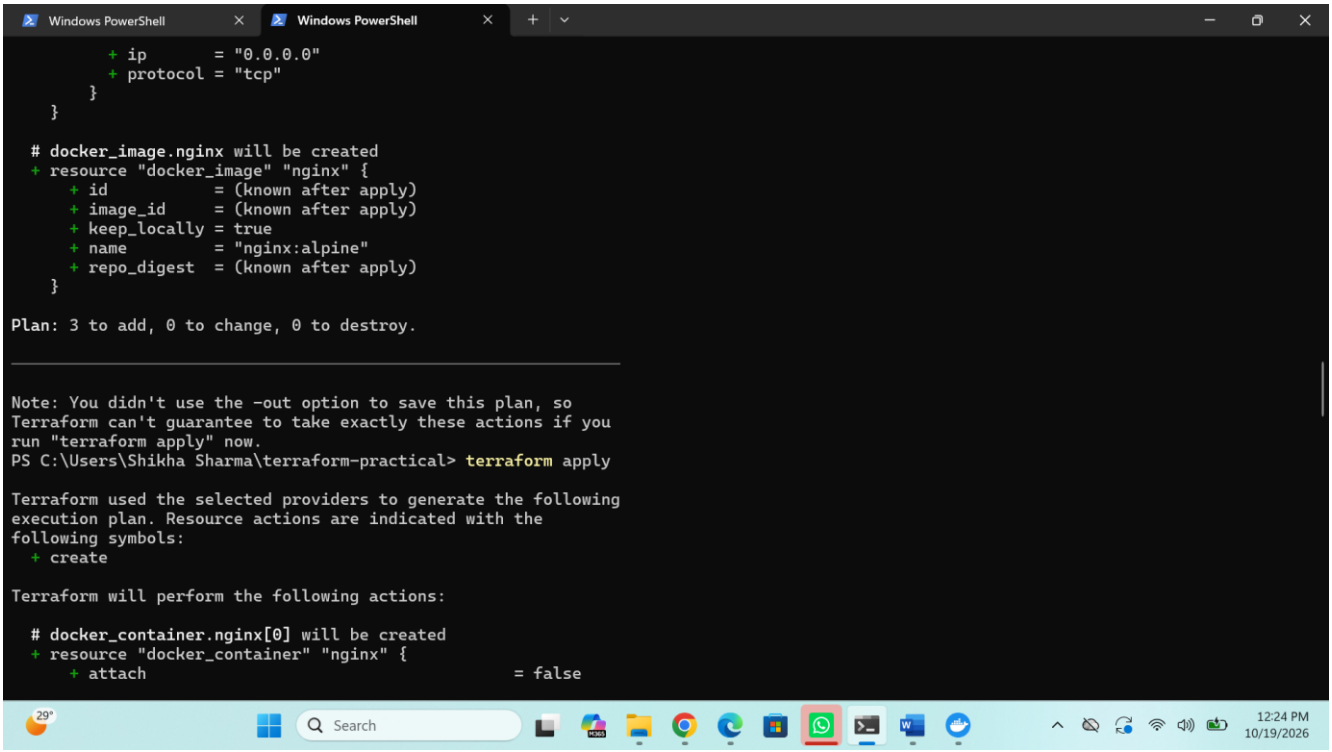
Terraform plan showing the remaining container and image resources.

Execution Screenshot 5



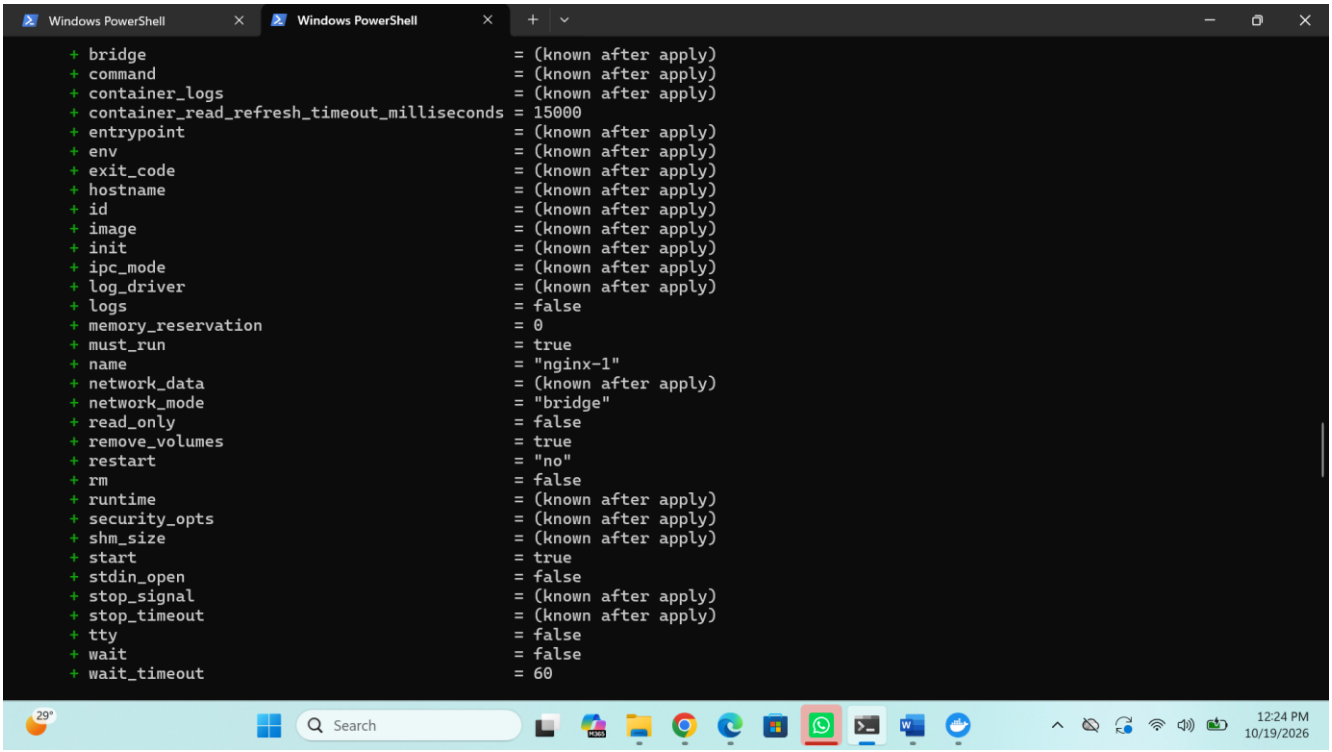
Terraform plan details and port mapping for nginx-2.

Execution Screenshot 6



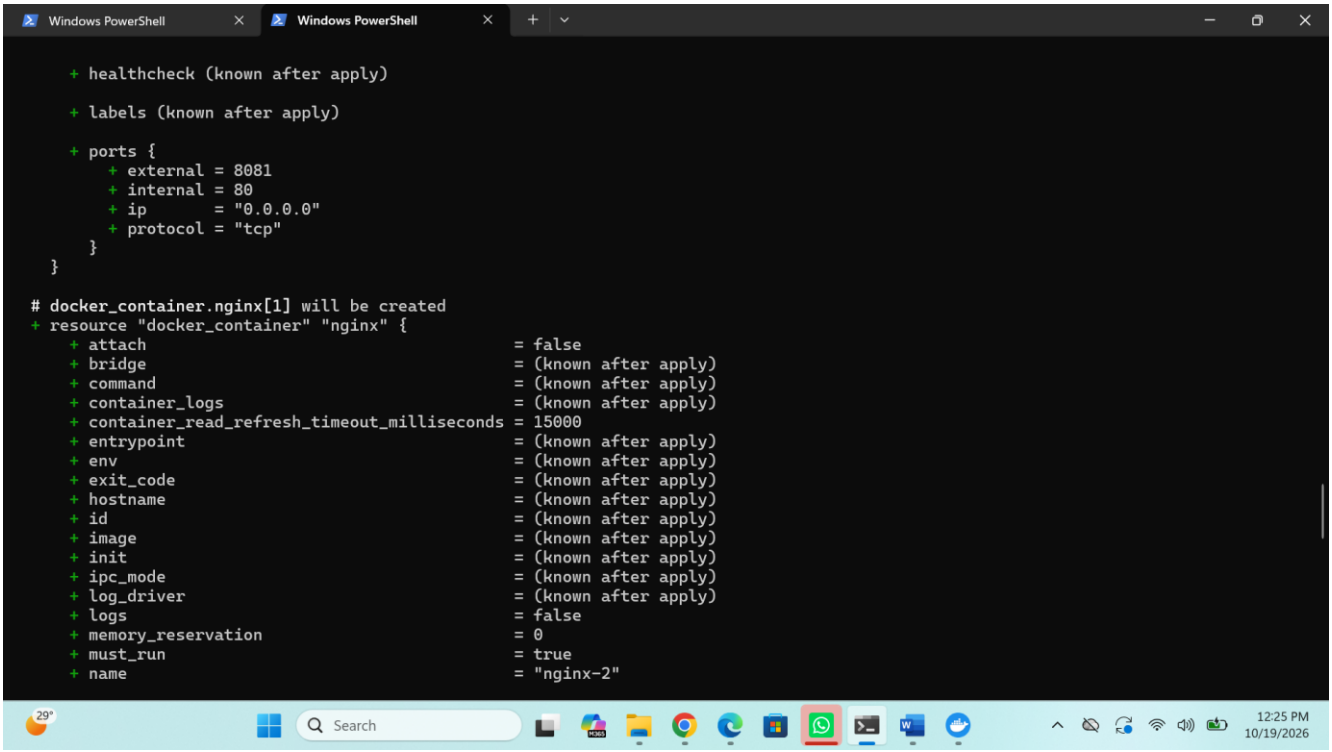
Plan summary and start of Terraform apply.

Execution Screenshot 7



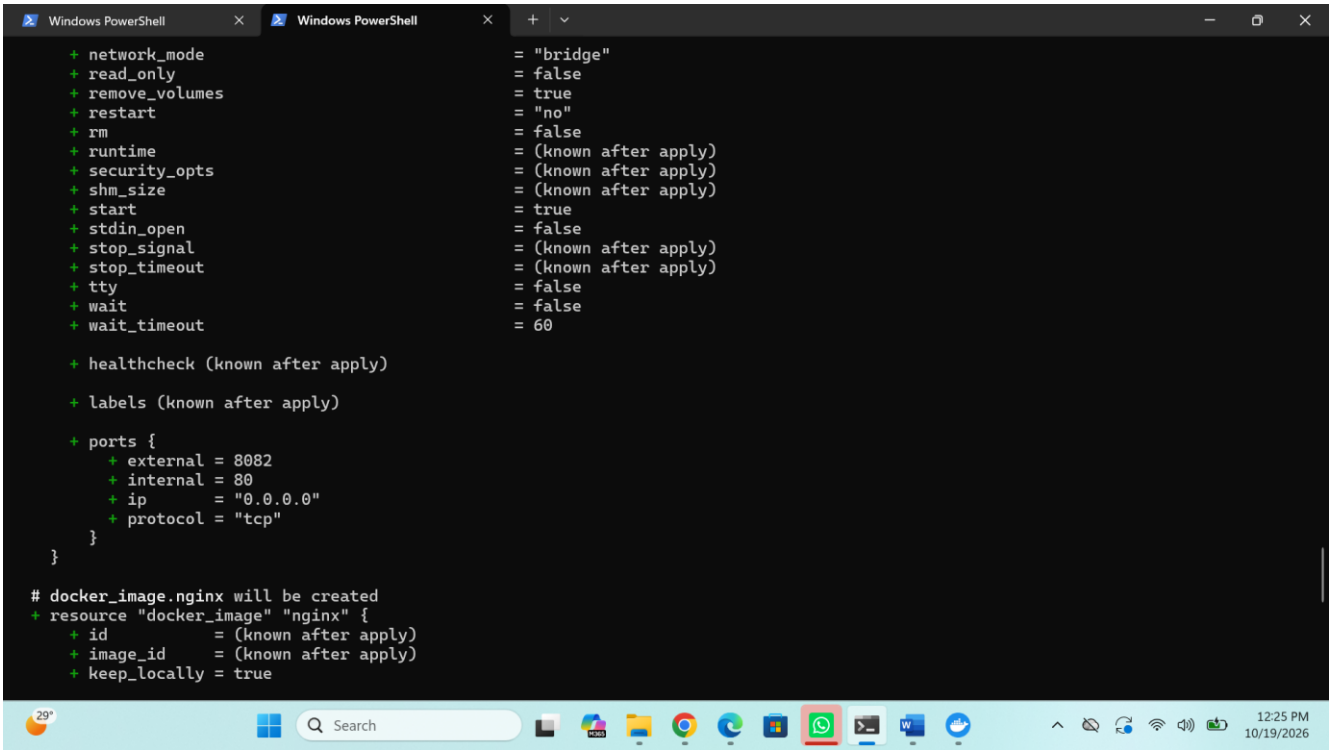
Terraform apply progress and resource creation.

Execution Screenshot 8



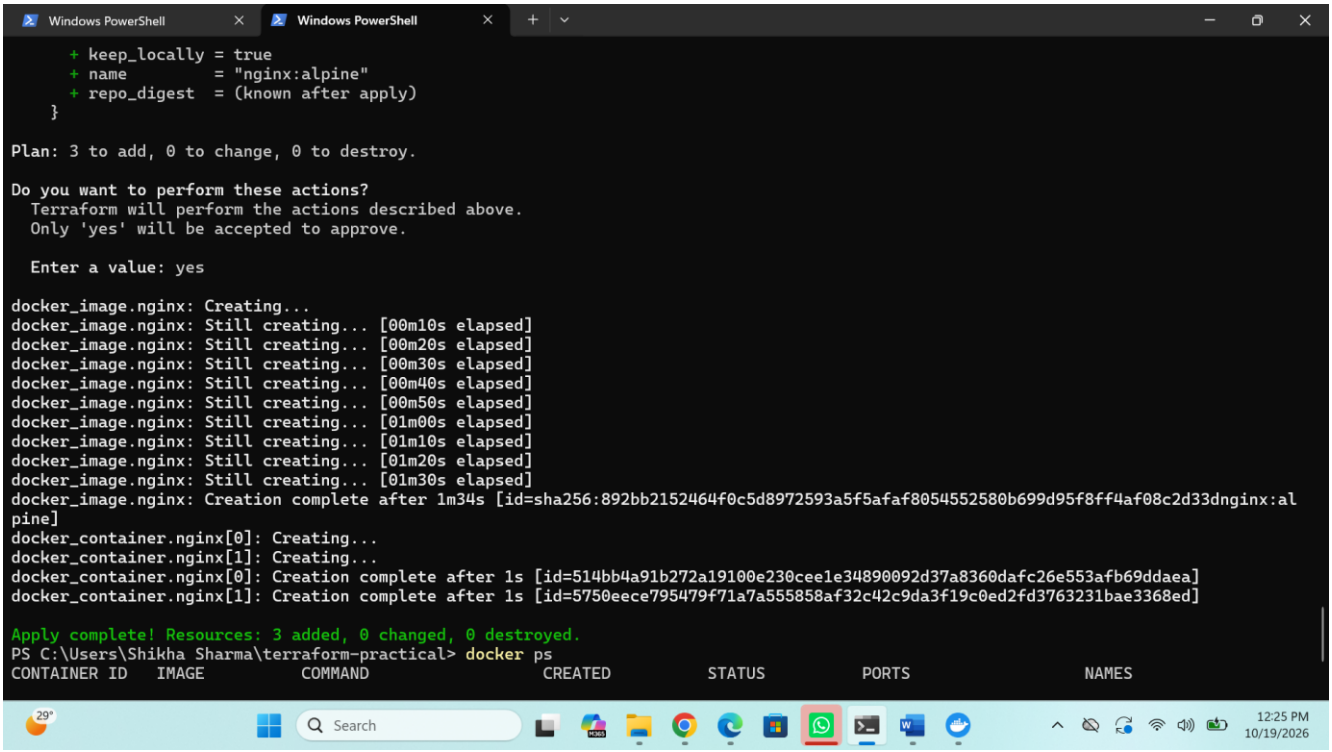
Terraform apply details for the container resources.

Execution Screenshot 9



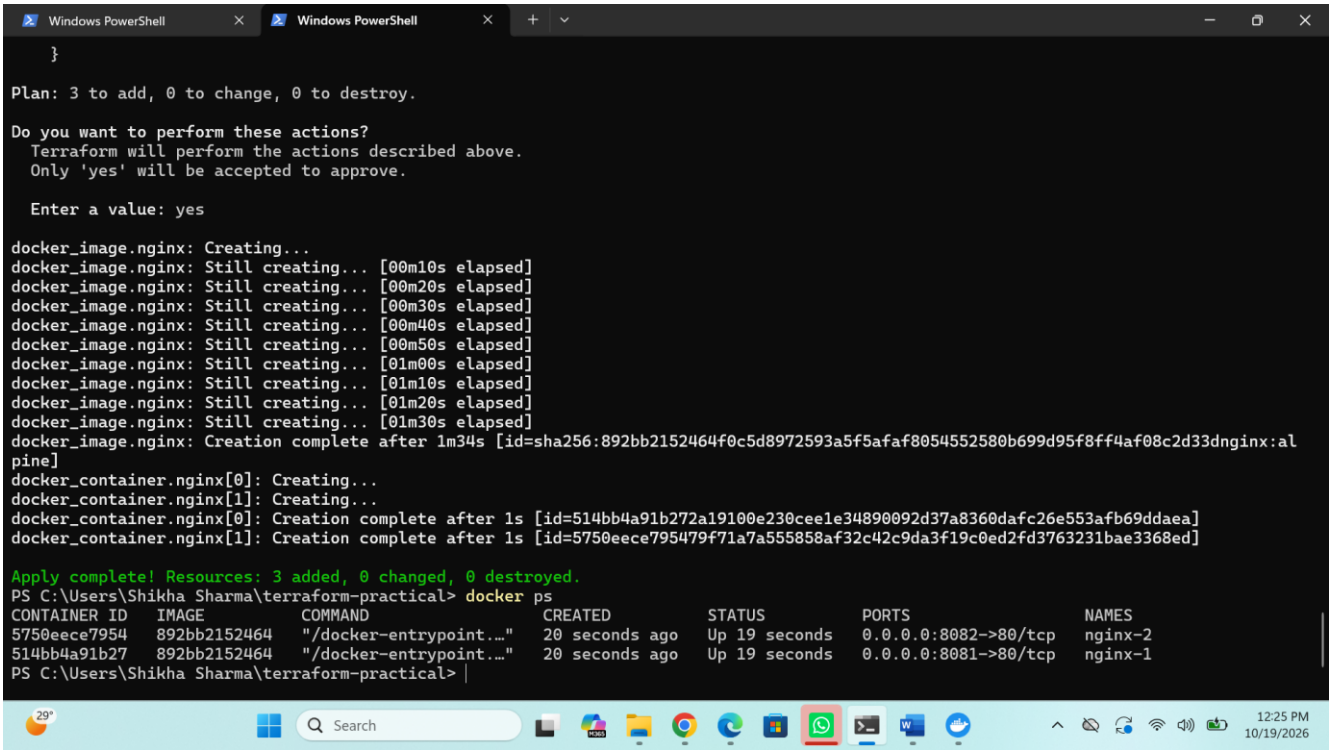
Terraform apply resource configuration and port mapping.

Execution Screenshot 10



Approval of the plan and successful image/container creation.

Execution Screenshot 11



Successful apply summary and docker ps verification.